

recover in the life span of the equipment. The real cost benefits of this system show up when you opt for a solar water heater for your society or building. Much greater economies of scale apply then. For an individual household the set up costs are around Rs. 130 per litre so a 100 litre setup would cost around Rs. 13,000 with an additional cost for installation. Here we're talking about evacuated tube solar collectors. Some of the things to look out for are insulating material quality and the reflective sheet underneath the tubes.

4 Buy star rated products

Energy Star Rating was first introduced by the US Department of Energy in 1992. The program ensures strict testing procedures in order to identify and promote energy efficient products. This reduces green house gases as well as saves money for the end users.

Specific to India, B.E.E (Bureau of Energy Efficiency, Govt. of India) has come out with a standard. Originally designed for Air Conditioning equipment, the objective of this labelling system is efficient use of energy and its conservation. BEE Star Label is now Mandatory – for Frost Free Refrigerators, Room ACs, Tubular Fluorescent Lamps and Distribution Transformers. We saw the BEE Star label on a couple of heaters too. The rating system is quite consumer friendly. The number of stars can vary from 1 to 5. Higher stars indicate higher energy efficiency and more energy bill savings for the customers. On a demo chart for refrigerators we noticed that a five star rated product consumed about half the electricity as non-rated refrigerator, which translated into a saving of Rs. 1750 per year.

5 Use natural resources

Many homes as well as offices are increasingly being designed to be naturally lit, cooled and heated. You too can structure your activities in such a way that use natural light more often. For instance if you want to read a book, there's no other source of light better than diffused sunlight. On the ventilation front, cross ventilation along with ceiling fans can do wonders. They're cheap too, costing less than 30 paise to run. Put up light reflecting films on windows to keep indoors cooler.

Clean air filters in ACs, this improves ventilation and air flow too. These methods should allow you to keep your cooling costs down by at least 10 per cent, which would be typically about Rs. 400 a year if your AC bill is Rs. 4000 a year.

6 Banish the vampires

Any electronic appliance when its on standby or even switched off draws some amount of power as long as it's plugged into a socket. This is known vampire power. While many of us know that such a thing occurs, we often underestimate the magnitude of how much this drain could be. Some estimates suggest that up to 22 % of all appliance consumption occurs from standby power drain. Adapters that usually don't come with a power button are the most hefty vampires. To avoid this it is recommended that you get power strips and flip just one switch for a set of connected appliances to turn them off. Or better yet, pull out the plug. The minimum reduction in bills is estimated to be around 10 per cent. So if you spend about Rs. 1000 on your electricity bill per month you should save Rs. 100.

7 Adjust the settings

- Turn your water heater thermostat down 2 degrees in winter and up several degrees in the summer.
- Set your A/C thermostat to 23°C – 25°C. It provides enough comfort and the AC operates for lesser time.
- Check if your refrigerator is unnecessarily cooling more than you need.
- Use appropriate load settings for your washing machine

8 Go LCD!

Some people believe that the lower power consumption of an LCD is overrated and that continuing with a CRT is the way to go. This is absolutely false. Let's just compare the power consumptions. A

Clean light

Dirty tube lights and bulbs reflect less light and can absorb 50 percent of the light; dust your tube lights and lamps regularly

How much can you save?

2 incandescents to LEDs	788
CRT to LCD	700
Solar water heating	1,500
Driving responsibly (30 km a day)	2,190
Pulling the plug	100
5 star energy refrigerator	1,750
Light and ventilation	400
Total	Rs 7,428

typical 19-inch CRT's power rating is set at around 100 W, and a similar-sized LCD, around 30 W. So what type of savings are we talking of here? Say you run a monitor for seven hours a day for a year. A CRT will then consume around 255 kilowatt-hours of power, and the LCD will pull around 76. Assuming that 1 unit (1 kilowatt-hour) of power is priced at around Rs 4, the difference in bills between the CRT and LCD over a year is around Rs 700. Multiply that by three years (the common warranty with most monitors), that's over Rs 2,100 saved. If you use your screen for more than seven hours, that would mean an even bigger saving. Now the cost to replace a CRT is also very affordable, since at the lower screen sizes LCDs are dirt cheap. Also remember, LCDs run at much lower temperatures. If you use an air conditioner in your computer room, the benefits of using an LCD might show up from further savings in this region as well. And of course let's not forget the glare reduction.

By now you would've realised that your little piggy bank of savings has become quite substantial. Although some of these calculations and assumptions are not exact, and others savings cannot be accurately measured, ball park figures suggest that you have enough savings to recover your annual Digit subscription. In fact there may possibly be enough savings to gift yourself another subscription! So go ahead go green – save the planet and save your money. **d**



Use this online energy calculator to figure out how much money it'll cost you to air condition your house and how much money you can save by opting for various star ratings. Head over to www.saveenergy.co.in. Use the simple and user friendly Energy Calculator that allows you to



choose different capacity of AC, quantity, efficiency level and operating time. The interface is drag and drop. The tool even shows a dummy layout of your house. Once you enter the relevant details, hit calculate. It will then tell you exactly how much your electricity cost will be for the year. Additionally it will inform you of any savings that can be availed by opting for higher star rating and how much.